ImpCML Coursework Instructions

Three documents make up the full course work for Mathematics for Machine Learning (60037/70015) that we are titling "ImpCML 2026". Due to Scientia file limits please find the following information at: [2026 repository link]

(1) - This instruction document

(2) - Coursework marking scheme and formatting

(3) - Kryptonite-*3.0* Mock Paper, Datasets and KryptoNet-40

This document gives instructions and expectations, the coursework formatting goes over formatting details for student write-ups (i.e., conference format) along with the mark scheme, and the prompt is a mock research paper. We describe each here.

1.) Coursework Instructions (This Document)

ImpCML (Imperial Coursework on Machine Learning) is meant to give students some practical experience with the tools we have seen throughout the course. The coursework objective is for groups of 3-4 students to read the Kryptonite-*3.0* mock paper, explore the proposed challenge datasets with a variety of methods (using valid experimental design), and write a response paper that soundly refutes the claim that Kryptonite-n is a dataset that cannot be solved with current ML methods, and that its foundation model, KryptoNet-40, cannot be adapted to it. Your paper should be complete with Abstract, Introduction, Methodology, Experimental Design, Experimental Results, and Discussion sections (with expectations for each outlined below).

The submitted content is comprised of two components: (A) a written report and (B) code to reproduce experiments. Expectations for these components are given below followed by a mark scheme.

1A) Written Report Expectations

The written report should be between 3-6 pages of content with unlimited additional space for references. Each report should have the following sections:

- **Abstract** - A short paragraph describing the main aims of the paper, method investigated for achieving that aim, and briefly mentioning the key results.
- **Introduction** - Sets a broader stage than the abstract. Spends a paragraph outlining the importance of the problem (in our case challenge datasets), a

paragraph describing how the paper seeks to solve the problem, a paragraph giving the reader some details of the solution structure (e.g., what experiments are carried out) and a paragraph summarizing the main takeaways of the experiments.

- **Methodology** - The mathematical foundations of any models used. This uses notation similar to what we have seen in class to orient the reader. E.g., if describing gradient descent writing it out formally and pointing out the hyper-parameters you are interested in like learning rate and number of epochs. This section must include your explanation of LoRA and its operation count (see 1C).
- **Experimental Design** - Typically in research papers this is left to the Appendix, but in this case we are asking students to dedicate a section of their paper to describing a hypothesis they seek to test, the experiment they will run to test this, and how they will ensure their results are valid (e.g., including error bars, different data splits).
- **Experimental Results** - Students will present the results of their experiments with tables or figures and describe how these plots validate or invalidate their hypothesis. *Importantly: text in figures should be no smaller than font size in the main text.*
- **Discussion** - Students will use this section to discuss what they were able to do, highlight any experiments they wish they could have done, and provide an environmental impact assessment of the experiments that they ran.

Together, each of these sections make up the full submission for the group's written report.

1B) Code and Labeling Expectations

In addition to the written report students must submit the following artifacts:

- **Code** to reproduce each of their figures. The code should be readable and interpretable to GTAs and myself. Ideally, students would create a script to reproduce each experiment individually. We expect that all plots that appear in the paper are accompanied by a Jupyter Notebook that reproduces it, pushed to the repository with its outputs.
- **Test labels** - Each challenge dataset comes with a set of features that does not have corresponding labels: hidden-kryptonite-20-X.npy, hidden-kryptonite-30-X.npy, hidden-kryptonite-40-X.npy and hidden-kryptonite-40-adapt-X.npy. Students are expected to submit a numpy

file for each of them: y_predicted_20.npy, y_predicted_30.npy, y_predicted_40.npy and y_predicted_40_adapt.npy. These .npy files should be in a directory at the top level of the repository titled "hiddenlabels". Each .npy should load into a binary vector of length 10000 and will be used to evaluate the final accuracy of each reported model. Run check_submission.py from the coursework repository to confirm the format before submitting. While students may explore several types of models in the context of their report, they are asked to only submit what they believe is their "best" predictions for each dataset.

- **Repository location** - State your GitLab repository URL and commit hash in a footnote on the first page of your report.

All experiments in this coursework run on a laptop CPU; no GPU is needed.

1C) Adapting KryptoNet-40 with LoRA (required)

The mock paper claims that its pretrained model, KryptoNet-40, cannot be adapted to Kryptonite-40-Adapt. Your report must:

1. Explain low-rank adaptation (LoRA) mathematically, in the notation of the course: for a layer with a $d_{\text{out}} \times d_{\text{in}}$ weight matrix W, what is trained, what is frozen, how A and B are initialised, and what role the rank r plays.
2. Count the operations needed for one training step, meaning one forward and one backward pass over a batch of N inputs, for a LoRA-adapted layer, and compare this with full fine-tuning of the same layer. Count multiplying an $n \times m$ matrix by an $m \times k$ matrix as nmk operations. Also give the number of trainable parameters in each case, and state how the count depends on the order in which the products are evaluated.
3. Apply LoRA to KryptoNet-40 and submit hidden-set predictions for Kryptonite-40-Adapt. You may implement LoRA yourselves or use an existing library such as peft; either way, your report must explain it.

Written document, GitLab Hash, and zip file containing labels must be submitted.

2.) Coursework Formatting

As this coursework aims to mimic an academic conference or workshop, we have modified the ICML formatting instructions to fit with this coursework. Students can duplicate the

provided Overleaf project in order to produce their write-up using LaTeX.

3.) Mock Paper

The mock paper "Kryptonite-*3.0*" lays out the perspective of challenge datasets as valid ways to audit machine learning models and puts forward a family of datasets that it claims proves machine learning is all hype: polynomial basis expansion, more training data and a purpose-built foundation model, KryptoNet-40, all fail on it, and the universal function approximation theorem does not really hold in practice. Additionally, this paper provides a github corresponding to the mock paper which contains the challenge datasets, the KryptoNet-40 weights, code to reproduce the described experiments e.g., polynomial regression, and the unlabelled grading data.